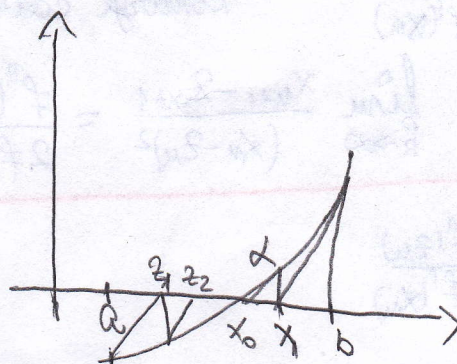


$$x_{m+1} = x_m - \frac{f(x_m)}{f'(x_m)}$$

$$[a, b], f(a)f(b) \leq 0, f'(x) > 0, f''(x) > 0$$



$$f - f(z_0) = f'(x_0)(x - z_0)$$

$$y = 0$$

$$-f(z_0) = f'(x_0)(x - z_0)$$

$$x - z_0 = -\frac{f(z_0)}{f'(x_0)}$$

$$z_1 = z_0 - \frac{f(z_0)}{f'(x_0)}$$

$$x_{m+1} - x_m = -\frac{f(x_m)}{f'(x_m)}$$

$$f(x_0) = f(b) > 0$$

$$m=0!$$

$$x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)} < 0$$

$$x_1 < x_0$$

$$f(x_{m+1}) > f(x_m) > 0$$

$$x_{m+1} - x_m < 0$$

$$x_{m+1} < x_m$$

$$f(x_{m+1}) < f(x_m)$$

$$m=1$$

$$x_2 - x_1 = -\frac{f(x_1)}{f'(x_1)} < 0$$

$$x_{m+1} = x_m - \frac{f(x_m)}{f'(x_m)} < x_m \Rightarrow x_{m+1} < x_m$$

Seuier:

$$z_{m+1} = z_m - \frac{f(z_m)}{f'(x_m)}$$

, x_m - rădăcină din metoda lui Newton

$$z_0 = a, x_0 = b$$

$$m=0: z_1 = z_0 - \frac{f(z_0)}{f'(x_0)}$$

→ arătăm că z_n și x_n crescătoare care tinde la rădăcină

$$z_0 = a, z_1 - z_0 = -\frac{f(z_0)}{f'(x_0)} > 0$$

$$z_{m+1} = z_m - \frac{f(z_m)}{f'(x_m)}$$

$$z_{m+1} - z_m = -\frac{f(z_m)}{f'(x_m)} > 0$$

□

I . Fie $f \in C^2[\alpha-\varepsilon, \alpha+\varepsilon]$, $f'(x) \neq 0$ și x_0, x_1 suficient de aproape de rădăcină (ε suficient de mic pt a fi verif cond din F. Newton), atunci:

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(x_n)}$$
 converge către unica soluție a ecuației $f(x)=0$
 și mai mult
$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - z_{n+1}}{(x_n - z_n)^2} = \frac{f''(\alpha)}{2f'(\alpha)}, f(\alpha)=0$$

dem
$$z_{n+1} = z_n - \frac{f(z_n)}{f'(x_n)}$$

$$\lim_{n \rightarrow \infty} z_n = \alpha$$

$$\alpha = \alpha - \frac{f(\alpha)}{f'(\alpha)} \Rightarrow f(\alpha) = 0.$$

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow z_{n+1} - x_{n+1} = z_n - x_n - \frac{f(z_n) - f(x_n)}{f'(x_n)}$$

$$f(z_n) = f(x_n) + (z_n - x_n)f'(x_n) + \frac{(z_n - x_n)^2 f''(c_n)}{2},$$

$$z_n < c_n < x_n$$

$$z_{n+1} - x_{n+1} = z_n - x_n - \frac{(z_n - x_n)f'(x_n)}{f'(x_n)} + \frac{(z_n - x_n)^2 f''(c_n)}{2} \quad | : (z_n - x_n)^2$$

$$\frac{z_{n+1} - x_{n+1}}{(z_n - x_n)^2} = \frac{f''(c_n)}{2f'(x_n)}$$

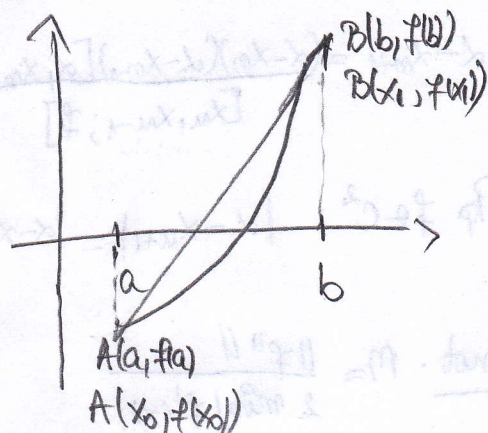
$$x_n \rightarrow \alpha$$

$$z_n < c_n < x_n \rightarrow c_n \rightarrow \alpha.$$

$$\lim_{n \rightarrow \infty} \frac{z_{n+1} - x_{n+1}}{(z_n - x_n)^2} = \frac{f''(\alpha)}{2f'(\alpha)}$$

$$|z_{n+1} - x_{n+1}| \leq M |z_n - x_n|^2 \Rightarrow (\text{ordinul de convergență către 0 e 2})$$

Metoda coordelor



$$y = \frac{x - x_0}{x_1 - x_0} f(x_1) + \frac{x_1 - x}{x_1 - x_0} f(x_0)$$

$$0 = x \left(\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right) - \frac{x_0 f(x_1) + x_1 f(x_0)}{x_1 - x_0}$$

$$x = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = \frac{x_0(f(x_1) - f(x_0)) + x_0 f(x_0) - x_1 f(x_0)}{f(x_1) - f(x_0)} =$$

$$x_2 = x_0 - \frac{f(x_0)}{\frac{f(x_1) - f(x_0)}{x_1 - x_0}}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{[x_{n-1}, x_n; f]}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{[x_{n-1}, x_n; f]}$$

metoda coordelor

(scriindu-l pe: $x = \frac{x_0 f(x_1) - x_1 (f(x_0) - f(x_1) - x_1 f(x_0))}{f(x_1) - f(x_0)}$)

[T] Dacă $x_0 = a$, $x_1 = b$ și dacă $f'(x) \neq 0$, $f''(x)$ nu schimbă semnul pe un intervalul $x \in [a - \varepsilon, a + \varepsilon]$ și dacă ε e suficient de mic, atunci metoda coordelor converge către unica soluție a ec $f(x) = 0$ și mai mult șirul x_n construit cu met coordelor are ordinul de convergență $n = \frac{1 + \sqrt{5}}{2}$

dem

$$a - x_{n+1} = a - x_n + \frac{f(x_n) - f(a)}{[x_n, x_{n-1}; f]}$$

$$a - x_{n+1} = a - x_n - (a - x_n) \frac{[a, x_n; f]}{[x_n, x_{n-1}; f]}$$

$$a - x_{n+1} = (a - x_n) \frac{([x_n, x_{n-1}; f]) - [a, x_n; f]}{[x_n, x_{n-1}; f]}$$

$$\alpha - x_{m+1} = \frac{(\alpha - x_m)(\alpha - x_{m-1})[f', x_m, x_{m-1}; f]}{[x_m, x_{m-1}; f]}$$

P. $f \in C^2$: $|\alpha - x_{m+1}| = |\alpha - x_m| |\alpha - x_{m-1}| \frac{|f''(\theta_m)|}{2|f'(x_m)|}$

not. $M = \frac{\|f''\|}{2 \min |f'(x)|}$

$$|\alpha - x_{m+1}| \leq |\alpha - x_m| |\alpha - x_{m-1}| M$$

$l_m = |\alpha - x_m| \rightarrow$ eroare absolută

$$l_{m+1} \leq M e_m l_{m-1} |M|$$

$$M l_{m+1} \leq M e_m \cdot M e_{m-1}$$

not $\delta_m = M e_m$

$$\delta_{m+1} \leq \delta_m \cdot \delta_{m-1}$$

$\boxed{m=1}$ $\delta_2 \leq \delta_1 \cdot \delta_0$

not $\delta = \max(\delta_1, \delta_0)$

$$\delta_2 \leq \delta^{1+1}$$

$\boxed{m=2}$ $\delta_3 \leq \delta_2 \cdot \delta_1 \leq \delta^2 \cdot \delta = \delta^{2+1}$

$$\delta_3 \leq \delta^3$$

$$\Rightarrow \delta_m \leq \delta^{f_{m-1} + f_{m-2}}$$

$$\delta_m \leq \delta^{f_m}$$

$f_m = f_{m-1} + f_{m-2} \rightarrow$ fibonacci
 $f_1 = f_0 = 1$

ec. caracteristică

$$r^2 - r - 1 = 0$$

$$\frac{1 \pm \sqrt{5}}{2} \quad f_m = A \left(\frac{1+\sqrt{5}}{2} \right)^m + B \left(\frac{1-\sqrt{5}}{2} \right)^m$$

$m=0$: $\begin{cases} A+B=1 \\ \frac{1}{2} + \frac{\sqrt{5}}{2} (A-B) = 1 \end{cases}$

$$\Rightarrow \begin{cases} A+B=1 \\ A-B=\frac{\sqrt{5}}{5} \end{cases}$$

$$A = \frac{5+\sqrt{5}}{10}$$

$$B = \frac{5-\sqrt{5}}{10}$$

$$f_m = \frac{\sqrt{5}}{5} \left(\frac{1+\sqrt{5}}{2} \right)^{m+1} + \frac{\sqrt{5}}{5} \left(\frac{1-\sqrt{5}}{2} \right)^{m+1}$$

$$|f_m| \leq \delta \cdot \frac{\sqrt{5}}{5} \left(\frac{1+\sqrt{5}}{2} \right)^{m+1} + \frac{\sqrt{5}}{5} \left(\frac{1-\sqrt{5}}{2} \right)^{m+1}$$

$$|f_m| \leq k \cdot \delta \frac{\sqrt{5}}{5} \left(\frac{1+\sqrt{5}}{2} \right)^{m+1} \rightarrow 0.$$

$$S_{m+1} \leq S_m S_{m-1}$$

$$\ln S_{m+2} - \ln S_m - \ln S_{m-1} = -\alpha_m^2$$

$$\text{not } C_n = \ln S_m$$

$$C_{m+1} - C_m - C_{m-1} = -\alpha_m^2$$

$$\text{for } r_1, r_2 \text{ root } r_1 = \frac{1+\sqrt{5}}{2}, r_2 = \frac{1-\sqrt{5}}{2}$$

$$C_{m+1} - (r_1 + r_2) C_m + r_1 r_2 C_{m-1} = -\alpha_m^2$$

$$C_{k+1} - r_1 C_k - r_2 (C_k - r_1 C_{k-1}) = -\alpha_k^2 \quad \left| \cdot \frac{1}{r_2^{k+1}} \right.$$

$$\frac{C_{k+1} - r_1 C_k}{r_2^{k+1}} - \frac{C_k - r_1 C_{k-1}}{r_2^k} = -\frac{\alpha_k^2}{r_2^{k+1}}$$

adunăm membre cu membre, $k=1, \dots, m$

$$\frac{C_{m+1} - r_1 C_m}{r_2^{m+1}} - \frac{C_1 - r_1 C_0}{r_2^1} = - \sum_{k=1}^m \frac{\alpha_k^2}{r_2^{k+1}} < 0$$

$$\frac{C_{m+1} - r_1 C_m}{r_2^{m+1}} < \frac{C_1 - r_1 C_0}{r_2} < 0$$

$$C_{m+1} < r_1 C_m$$

$$\ln |\alpha - x_{m+1}| \leq r_1 \ln |\alpha - x_m| + M$$

$$|\alpha - x_{m+1}| \leq |\alpha - x_m|^{r_1} M$$

$$|\alpha - x_{m+1}| \leq M |\alpha - x_m|^{\frac{1+\sqrt{5}}{2}} \Rightarrow \text{ordul de convergență e } \frac{1+\sqrt{5}}{2}.$$

METODE NUMERICE DE REZOLVARE A ECUATIILOR DIFERENTIALE

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$$

$$f(x, y) = (f_1(x, y_1, y_2, \dots, y_n), f_2(x, y_1, y_2, \dots, y_n), \dots, f_n(x, y_1, y_2, \dots, y_n))^T$$

$$y_1' = f_1(x, y_1, \dots, y_n)$$

$$\vdots$$

$$y_n' = f_n(x, y_1, \dots, y_n)$$

$$y(x_0) = y_0: \begin{cases} y_1(x_0) = y_1^0 \\ y_2(x_0) = y_2^0 \\ \vdots \\ y_n(x_0) = y_n^0 \end{cases}$$

$$\boxed{\times} \quad y''' + a(x)y' + b(x)y = g(x)$$

$$y''' = -a(x)y' - b(x)y - g(x)$$

$$y' = z$$

$$y'' = z' = z_1$$

$$y''' = z_1' = -a(x)z - b(x)y - g(x)$$

$$\begin{cases} y' = z \\ y'' = z_1 \end{cases}$$

$$y''' = -a(x)z - b(x)y - g(x)$$

$\square$ Fie $f: I \times D \rightarrow \mathbb{R}$ continuă cu proprietate că este Lipschitziană în raport cu variabila y , adică $\exists k > 0$, a.i. $\forall y_1, y_2 \in D, \forall x \in I$ să avem:

$$|f(x, y_1) - f(x, y_2)| \leq k |y_1 - y_2|$$

dacă fct vectorială: $\|f(x, y_1) - f(x, y_2)\| \leq k \|y_1 - y_2\|$

$$f = (f_1, f_2, \dots, f_n)^T: \sqrt{\sum_{i=1}^n (f_i(x, y_1^1, y_2^1, \dots, y_n^1) - f_i(x, y_1^2, y_2^2, \dots, y_n^2))^2} \leq k \sqrt{\sum_{i=1}^n (y_i^1 - y_i^2)^2}$$

$$\exists \alpha > 0, y: [\alpha - x_0, \alpha + x_0] \rightarrow \mathbb{R}$$

$$y(x_0) = y_0$$

$$y' = f(x, y)$$

$$\boxed{\text{ex}} \begin{cases} y' = y + g(x) \\ y(x_0) = y_0 \end{cases}$$

$$f(x, y) = y + g(x) \Rightarrow \frac{\partial f}{\partial y} = 1.$$

arătăm că e lipschitziană

$$|f(x, y_1) - f(x, y_2)| = |y_1 - y_2| \left| \frac{\partial f}{\partial y}(x, \theta) \right| = |y_1 - y_2| \leq M |y_1 - y_2|$$

$\Downarrow$ lipschitziană $\Rightarrow$ soluție